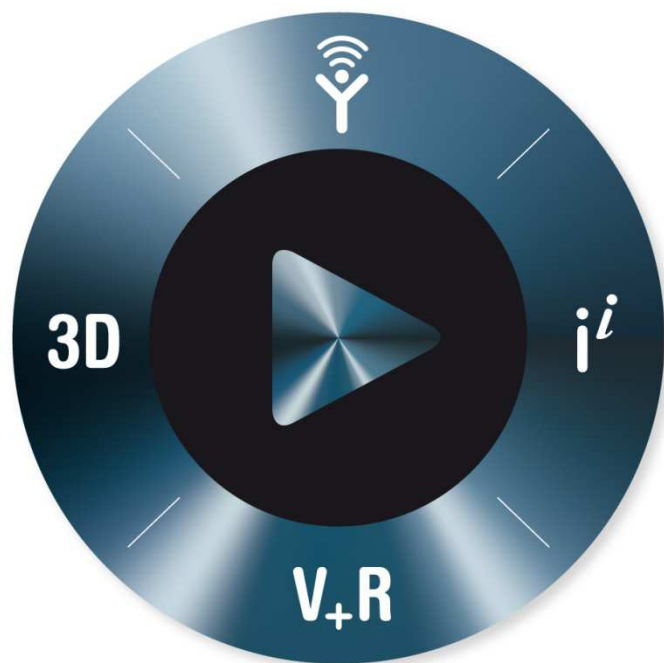




3DEXPERIENCE

Abaqus真实仿真技术 在兵器行业中的应用

赵友选

yzb@3ds.com

18611685690

2012-11-15

Agenda

- ▶ Abaqus软件简介
- ▶ Abaqus真实仿真技术在兵器行业中的应用案例

Abaqus产品介绍

Abaqus/CAE

- ▶ 为Abaqus求解器提供快速交互式的前后处理环境
- ▶ Abaqus的建模、分析、监测和控制、以及结果评估的完整界面

Abaqus/Standard

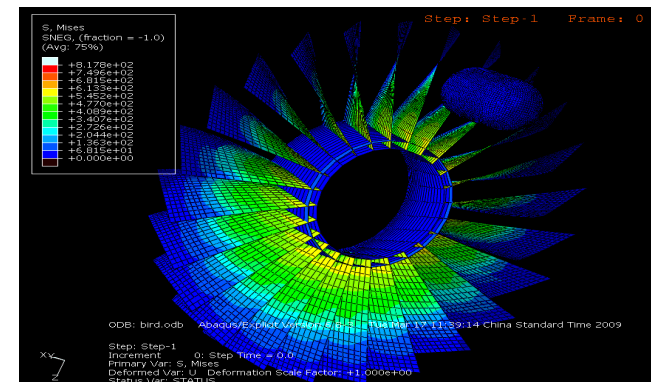
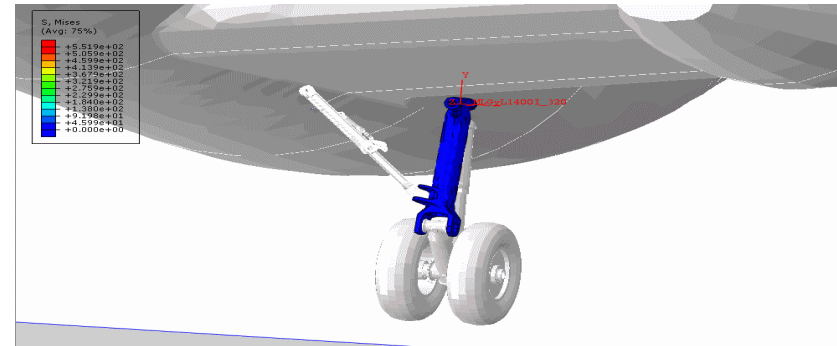
- ▶ 主要用于结构静态、动态线性和非线性分析
- ▶ 耦合分析

Abaqus/Explicit

- ▶ 瞬态的大变形和高度非线性分析
- ▶ 可以在Abaqus/Standard分析结束状态进行继续分析

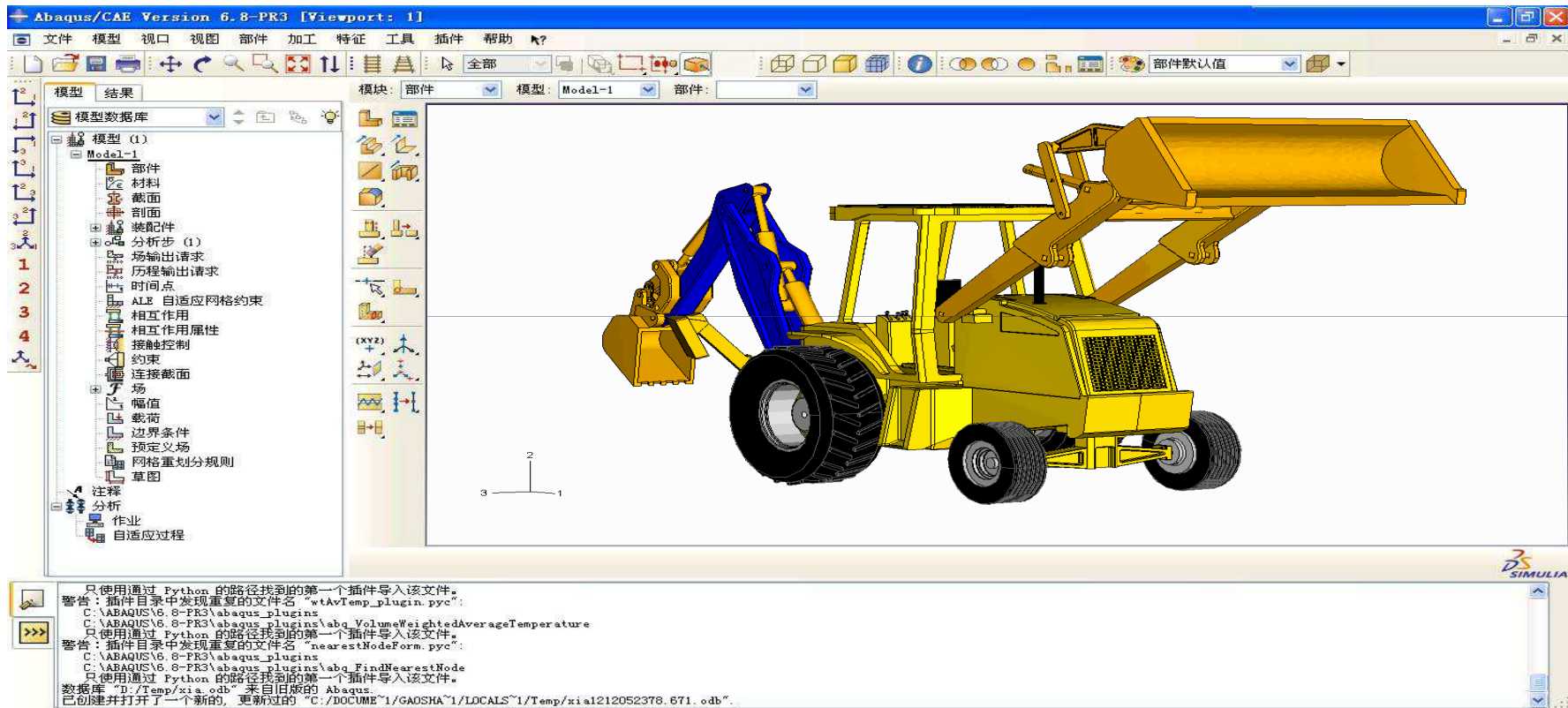
Abaqus/CFD

- ▶ 求解流体流动和传热问题
- ▶ 结合Abaqus/Standard和Abaqus/Explicit进行流固耦合(FSI)分析



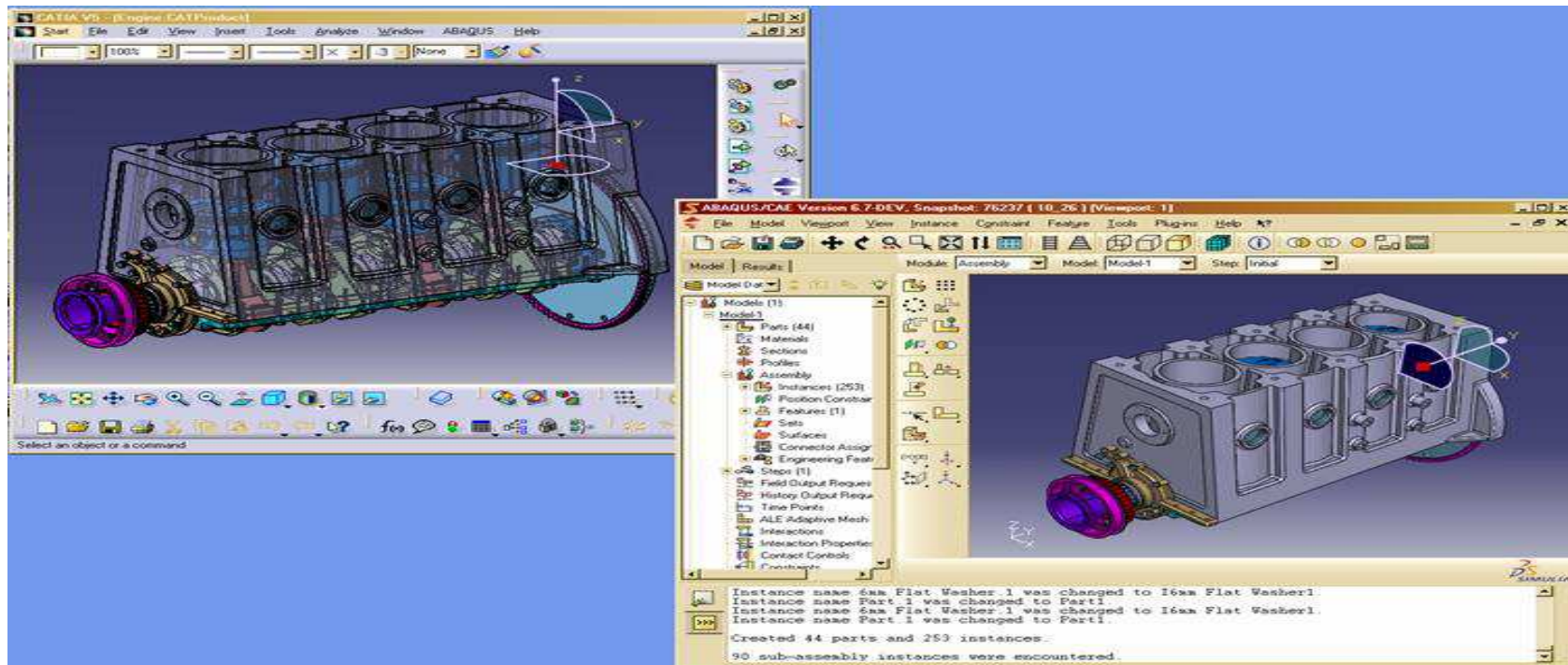
Abaqus 6.8EF版正式推出中文汉化包

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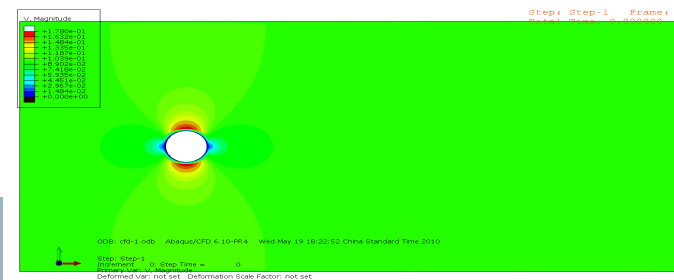
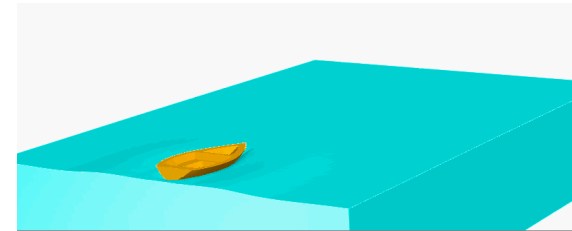
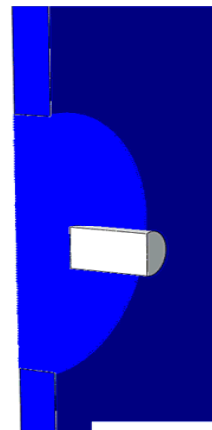
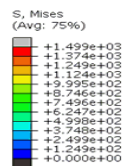
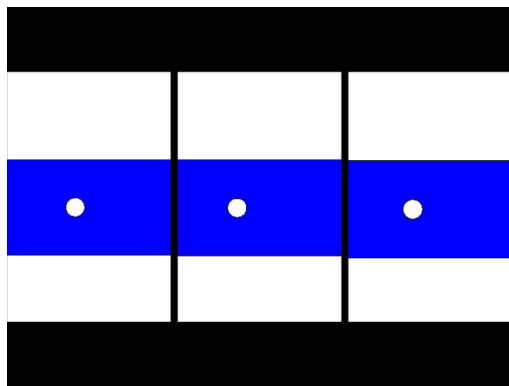
Abaqus重要的新功能

- 实现和CATIA、ProE、Solidwork以及NX的协同建模



Abaqus重要的新功能

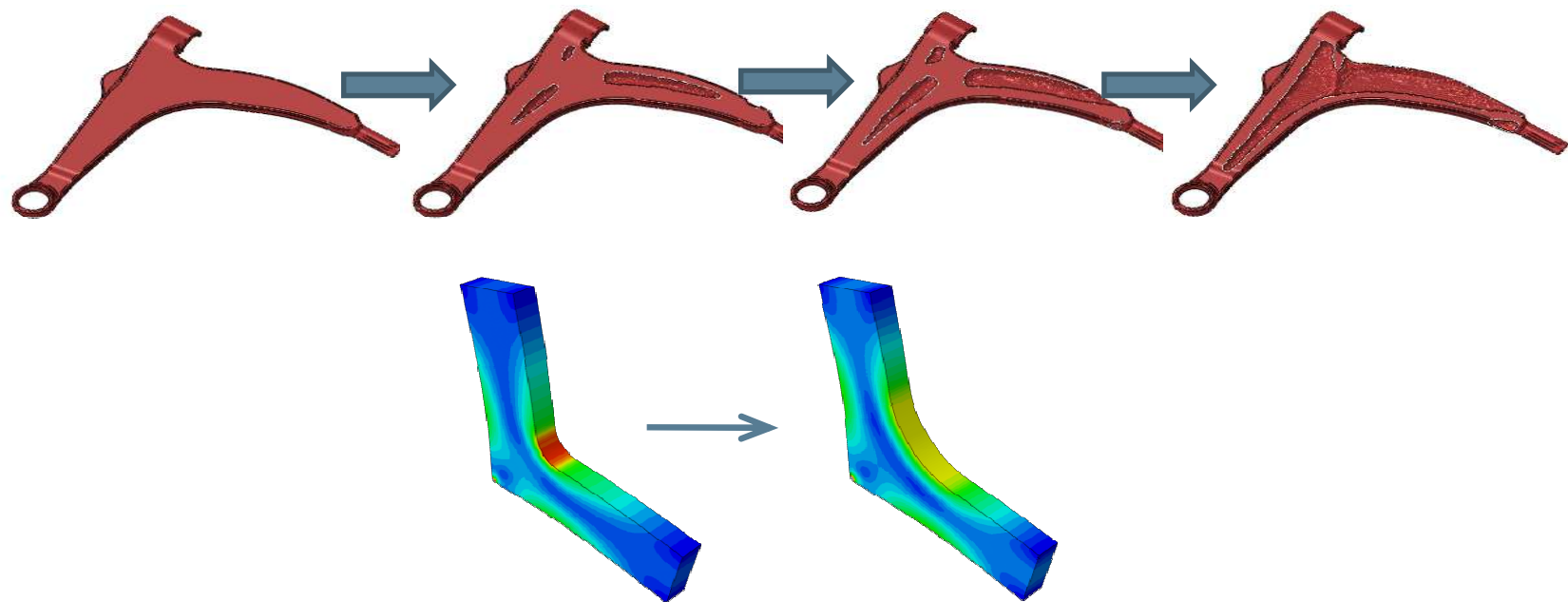
- ▶ Standard通用接触
- ▶ 全新的CEL求解器
- ▶ XFEM
- ▶ CFD
- ▶ SPH



ask the right questions
change the world.

Abaqus重要的新功能

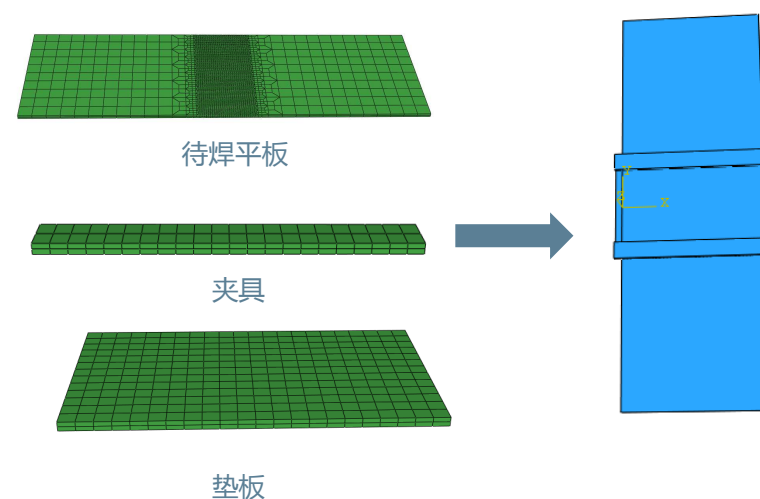
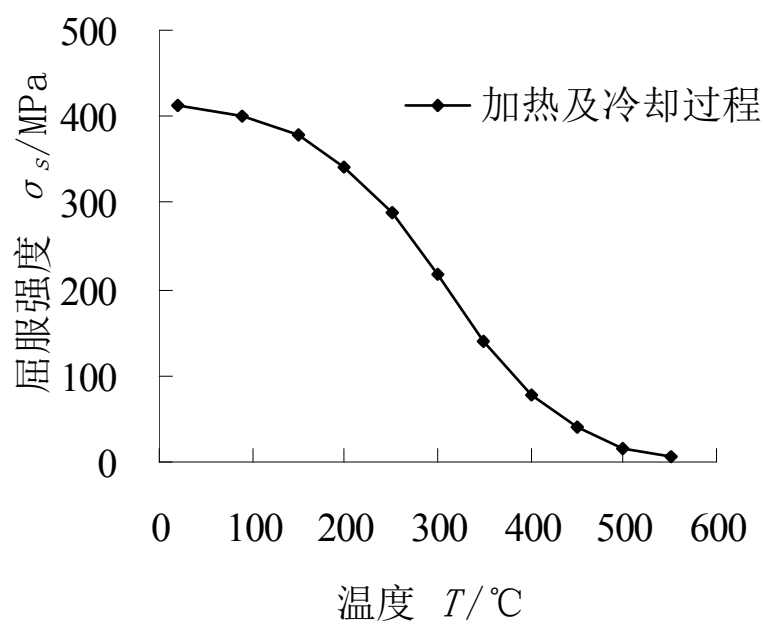
► ATOM



Abaqus真实仿真技术在兵器行业中的应用案例

铝合金焊接

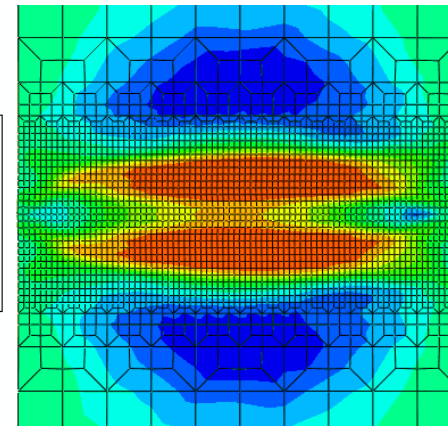
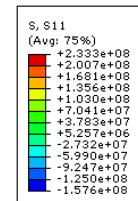
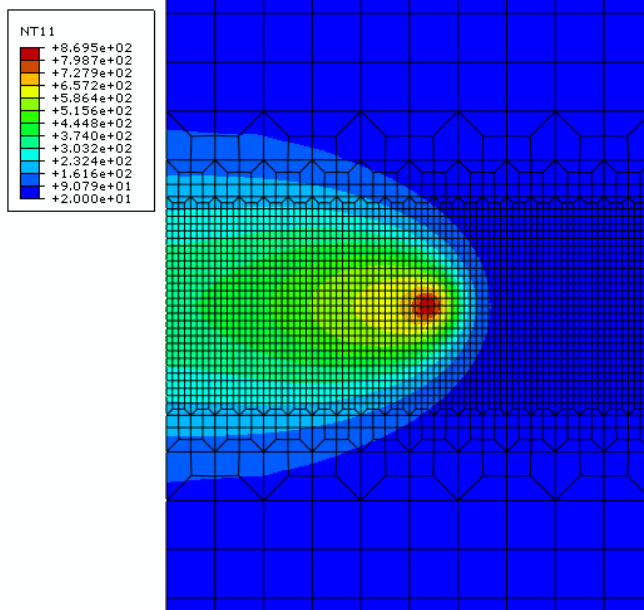
- ▶ 待焊平板材料为铝合金6056，材料本构参数与温度相关



铝合金焊接

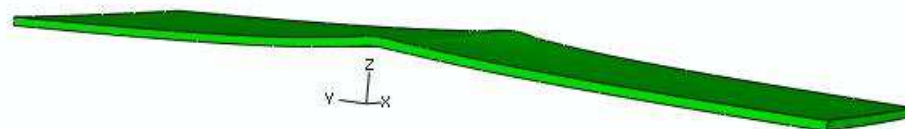
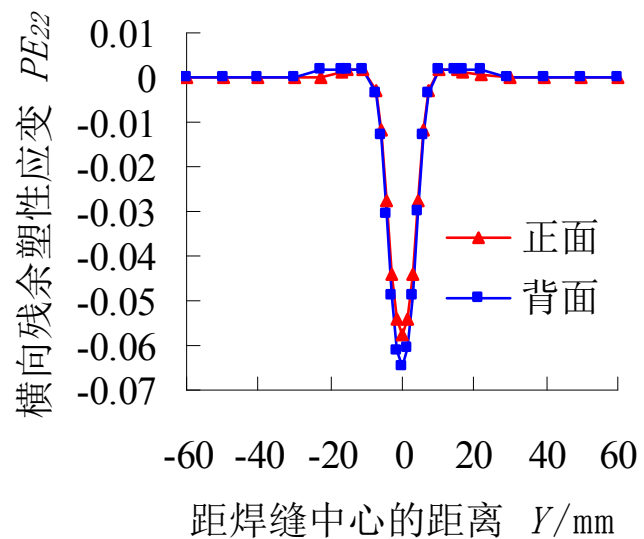
► 移动高斯热源分布公式

$$q(r) = \frac{3Q}{\pi r_H^2} \exp\left(-\frac{3r^2}{2r_H^2}\right)$$



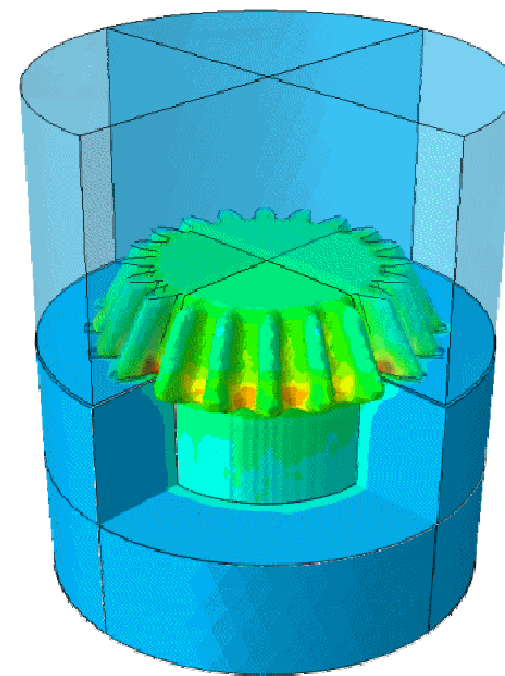
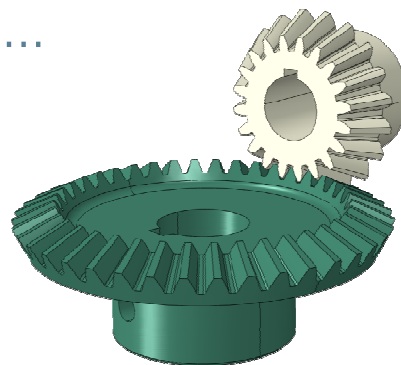
铝合金焊接

► 焊后残余变形趋势与实验结果对照



齿轮热成型

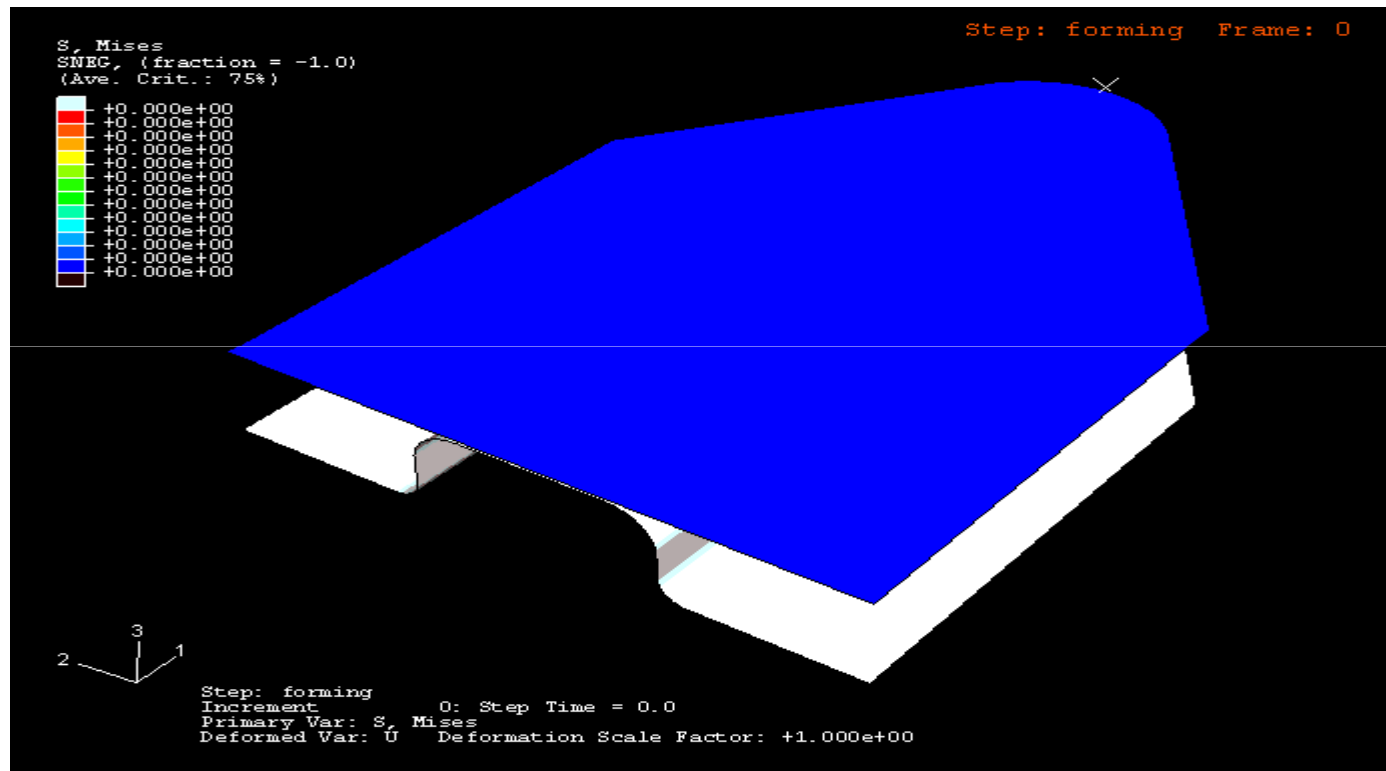
- ▶ Extended CEL to thermal stress simulation with contact interactions
 - ▷ Complements thermal stress element introduced in Abaqus 6.11
- ▶ Applications include forging, mold filling, manufacturing processes, ...



Hot forging

零件加工成型

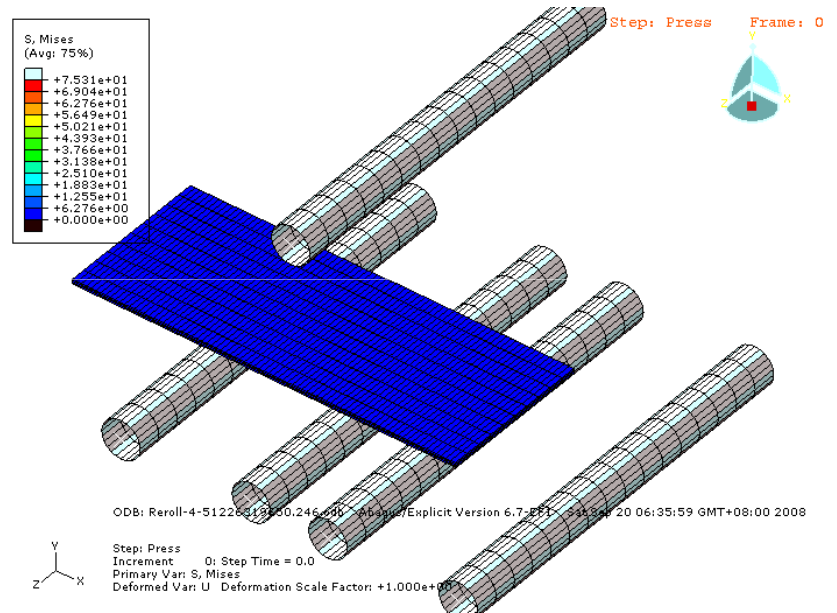
钣金成型之冲压成型模拟



零件加工成型

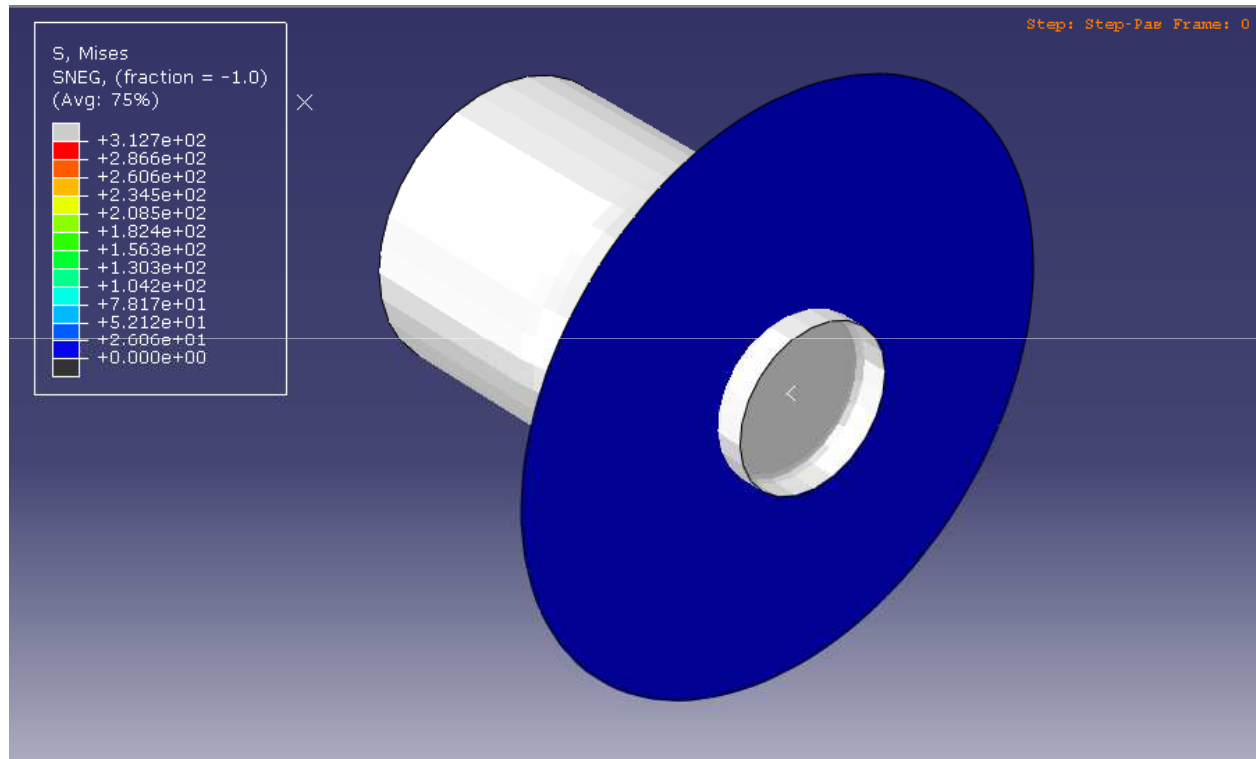
滚弯成型

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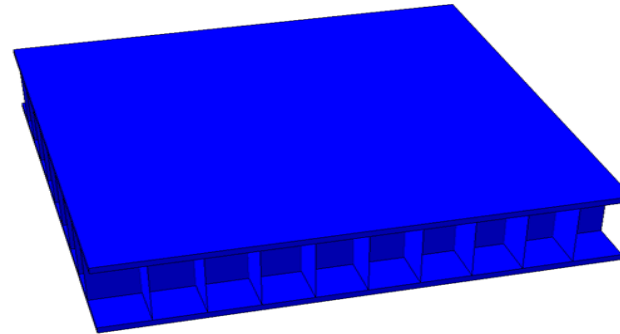
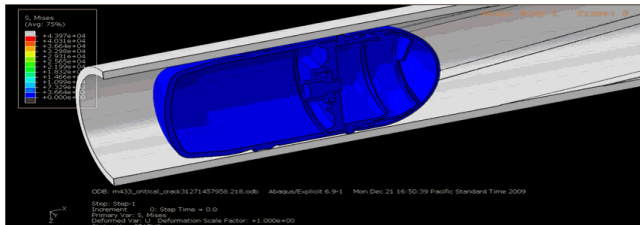
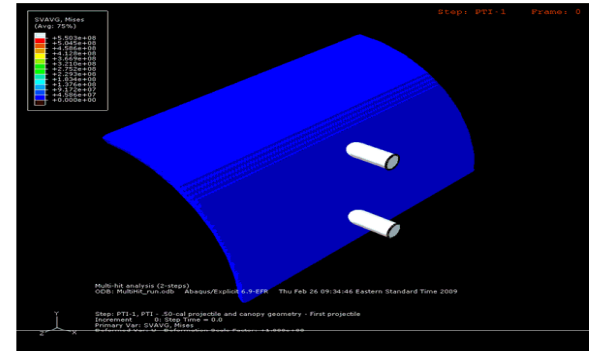
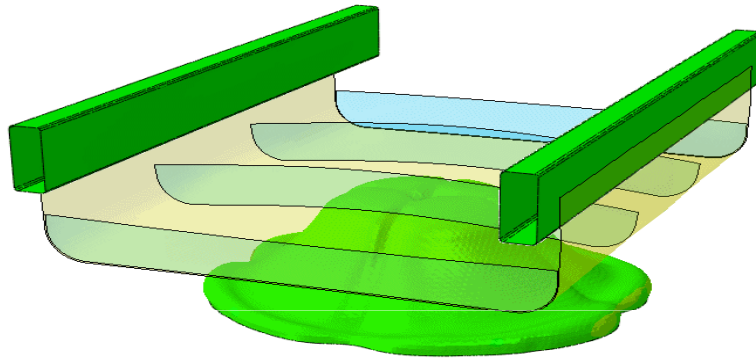


零件加工成型

旋压成型



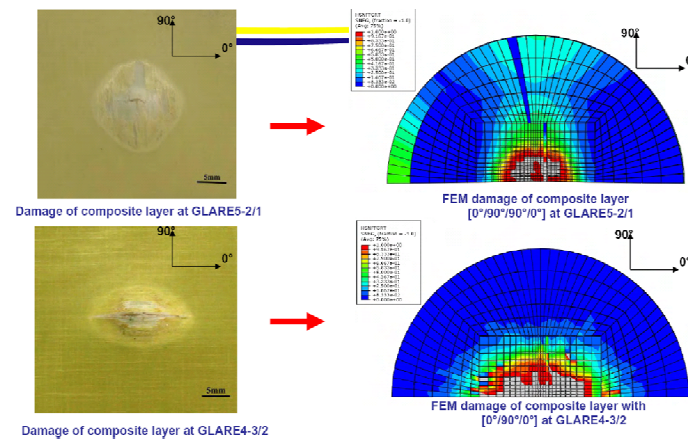
国防行业中的部分应用



Ask the right questions
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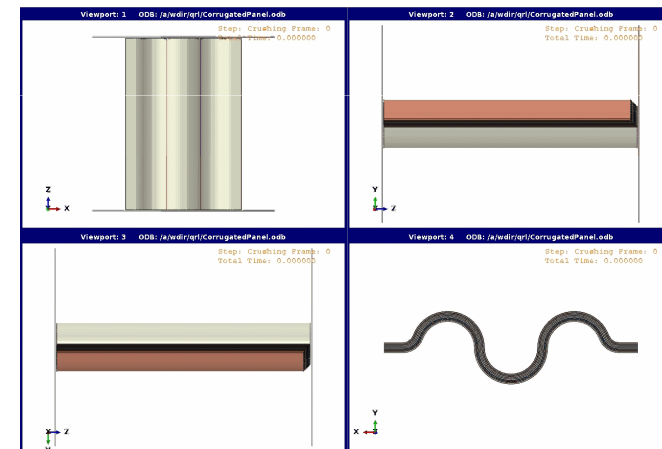
Composites Fracture & Failure

- VCCT, Damage modeling, Hashin, Puck, Cohesive for Delamination



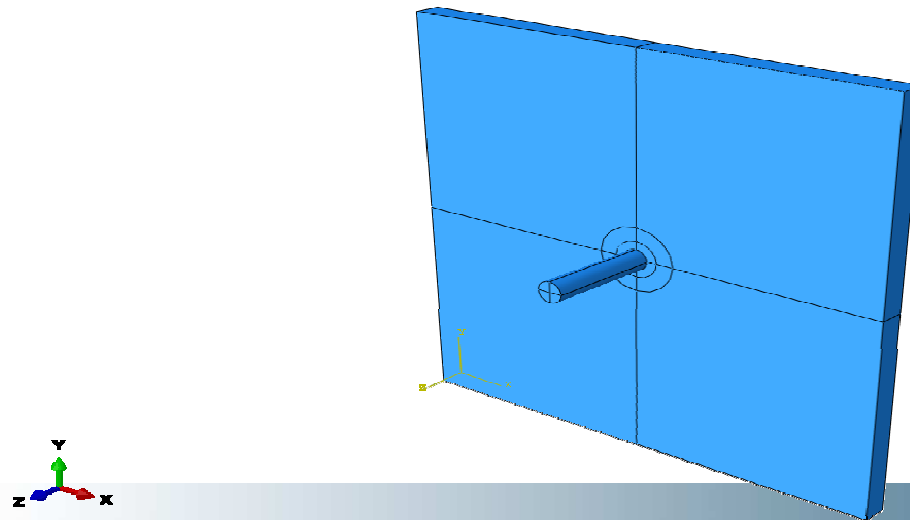
BVID using Hashin's criteria

Composites crush using continuum shells, Puck's criteria, cohesive surface for delamination, and general contact



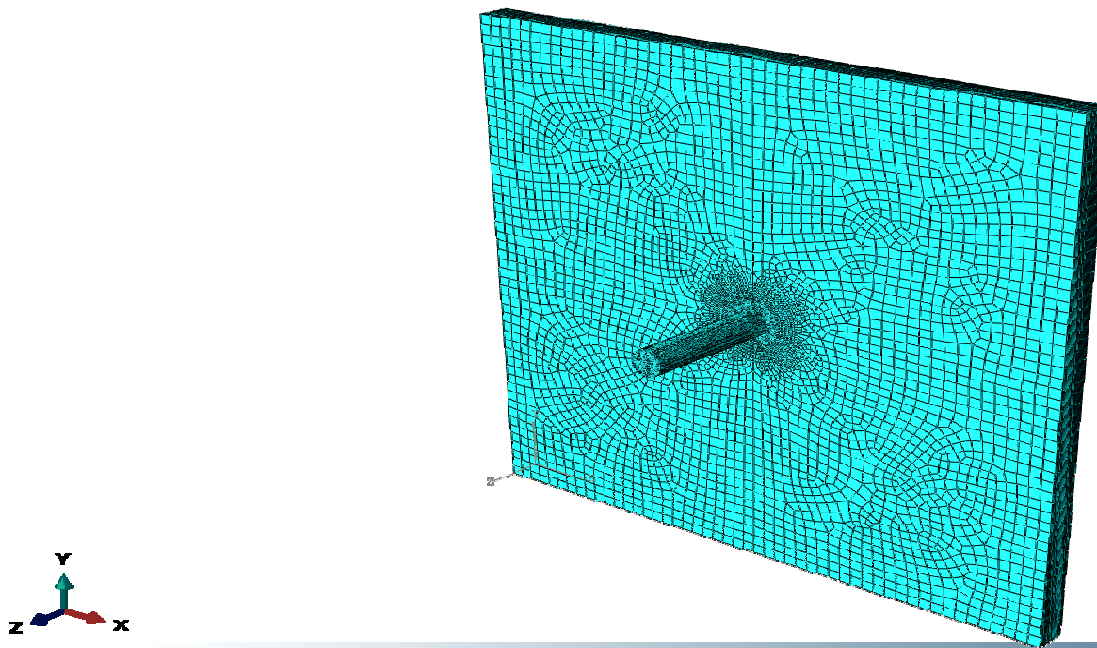
Abaqus模拟A3钢钝头弹撞击45钢

1. 分别以250m/s、350m/s、550m/s、750m/s的速度进行撞击
2. 采用J-C塑性以及损伤失效本构
3. 得到在不同撞击速度下的弹体失效模式



Abaqus模拟A3钢钝头弹撞击45钢

有限元网格：



Abaqus模拟A3钢钝头弹撞击45钢

*Material, name=A3

*Damage Initiation, criterion=JOHNSON COOK

0.3, 0.9, 2.8, 0., 0., 1795., 300., 1.

*Damage Evolution, type=DISPLACEMENT

0.,

*Density

7800.,

*Elastic

2e+11, 0.3

*Inelastic Heat Fraction

1.,

*Plastic, hardening=JOHNSON COOK

4.1e+08, 2e+07, 0.08, 0.55, 1795., 300.

*Rate Dependent, type=JOHNSON COOK

0.1, 1.

*Specific Heat

469.,

*Material, name=steel45

*Damage Initiation, criterion=JOHNSON COOK

0.1, 0.76, 1.57, 0.005, -0.84, 1795., 300., 1.

*Damage Evolution, type=DISPLACEMENT

0.,

*Density

7800.,

*Elastic

2e+11, 0.3

*Inelastic Heat Fraction

1.,

*Plastic, hardening=JOHNSON COOK

5.06e+08, 3.2e+08, 0.28, 1.06, 1795., 300.

*Rate Dependent, type=JOHNSON COOK

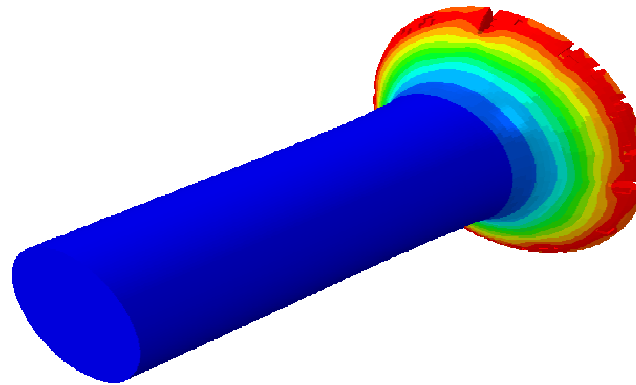
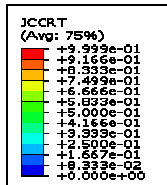
0.064, 1.

*Specific Heat

469.,

Abaqus模拟A3钢钝头弹撞击45钢

•以250m/s速度撞击



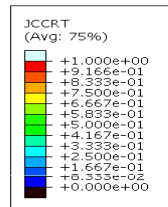
ODB: a250.odb Abaqus/Explicit 6.10-2 Sun Dec 19 05:43:49 GMT+08:00 2010

Step: Step-1
Increment: 8311; Step Time = 2.0001E-04
Primary Var: JCCRT
Deformed Var: U Deformation Scale Factor: +1.000e+00
Status Var: STATUS

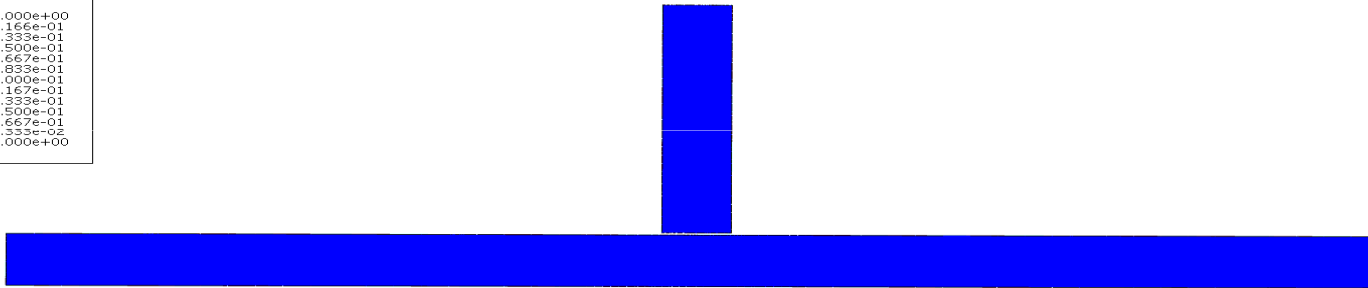
200 μ s时刻

Abaqus模拟A3钢钝头弹撞击45钢

•以250m/s速度撞击



Step: Step-1 Frame: 0
Total Time: 0.000000

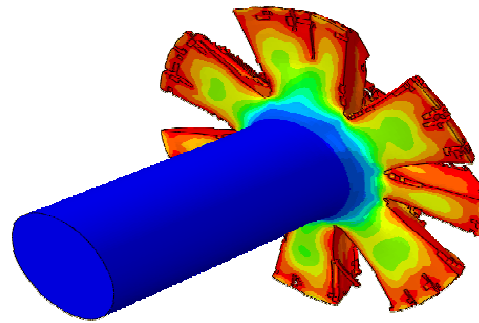
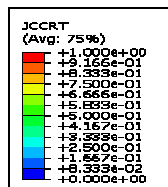


ODB: a250.odb Abaqus/Explicit 6.10-2 Sun Dec 19 05:43:49 GMT+08:00 2010

Step: Step-1
Increment: 0; Step Time = 0.0
Primary Var: JCCRT
Deformed Var: U Deformation Scale Factor: +1.000e+00
Status Var: STATUS

Abaqus模拟A3钢钝头弹撞击45钢

•以350m/s速度撞击



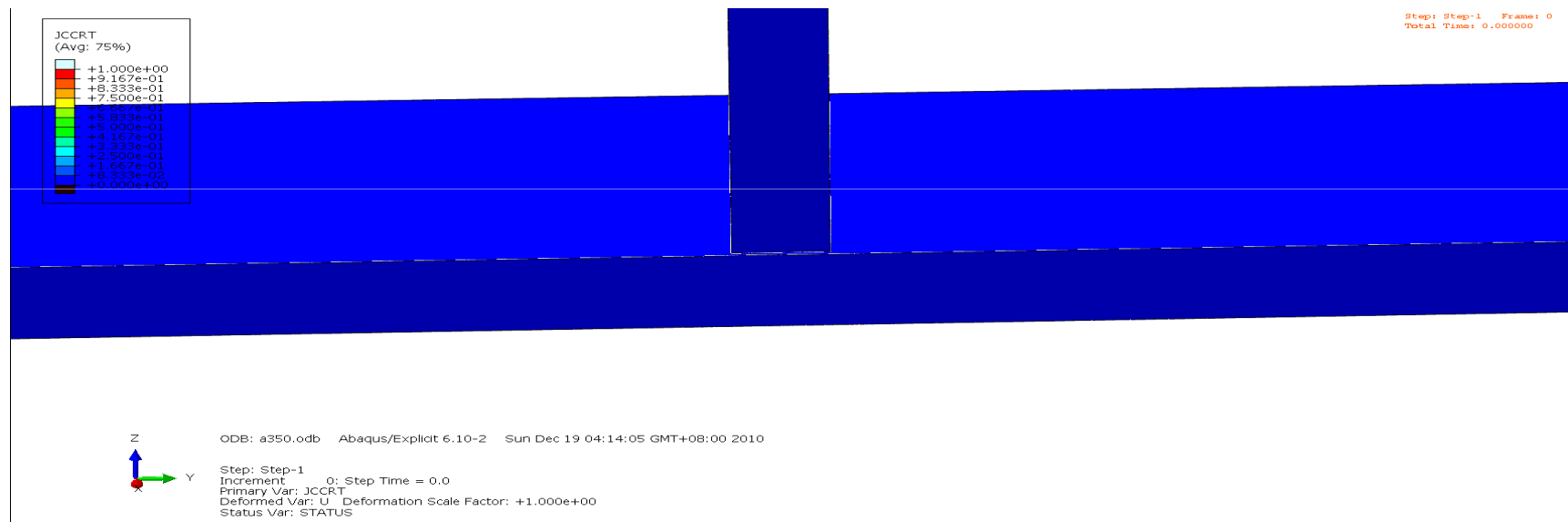
ODB: s350.odb Abaqus/Explicit: 6.10-2 Sun Dec 19 04:14:05 GMT+08:00 2010

Step: Step-1
Increment: 8983; Step Time = 2.0001E-04
Primary Var: JCCRT
Deformed Var: U Deformation Scale Factor: +1.000e+00
Status Var: STATUS

200 μ s时刻

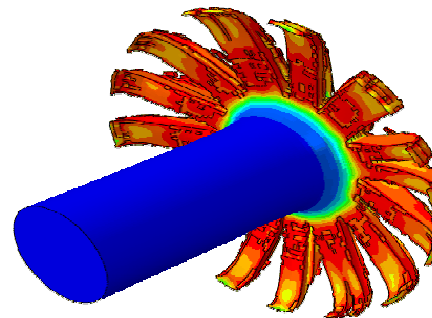
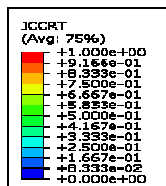
Abaqus模拟A3钢钝头弹撞击45钢

•以350m/s速度撞击



Abaqus模拟A3钢钝头弹撞击45钢

- 以550m/s速度撞击



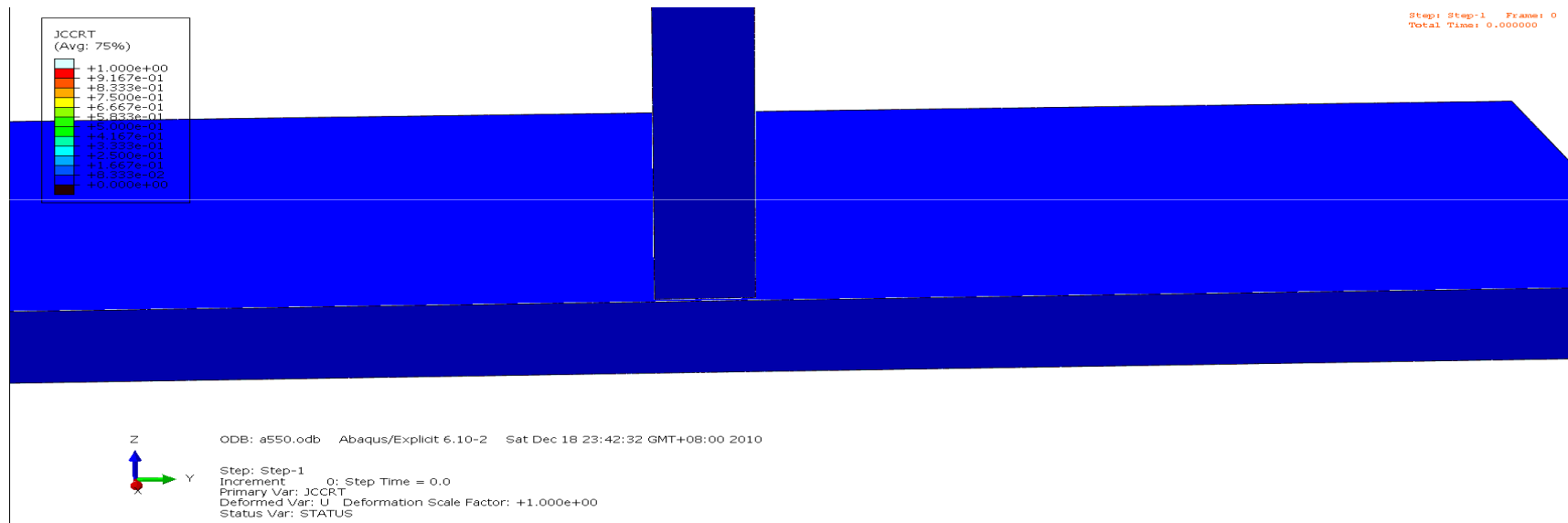
ODB: e550.odb Abaqus/Explicit: 6.10-2 Sat Dec 18 23:42:32 GMT+08:00 2010

Step: Step-1
Increment: 4248; Step Time = 1.0000E-04
Primary Var: JCCRT
Deformed Var: U Deformation Scale Factor: +1.000e+00
Status Var: STATUS

100 μ s时刻

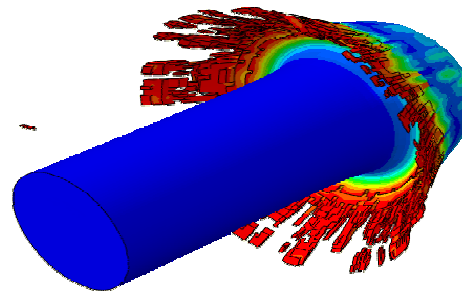
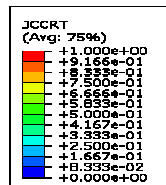
Abaqus模拟A3钢钝头弹撞击45钢

•以550m/s速度撞击



Abaqus模拟A3钢钝头弹撞击45钢

- 以750m/s速度撞击



ODB: a750.odb Abaqus/Explicit 6.10-2 Sun Dec 19 01:34:20 GMT+08:00 2010

Step: Step-1
Increment 3407: Step Time = 7.0005E-05
Primary Var: JCCRT
Deformed Var: U Deformation Scale Factor: +1.000e+00
Status Var: STATUS

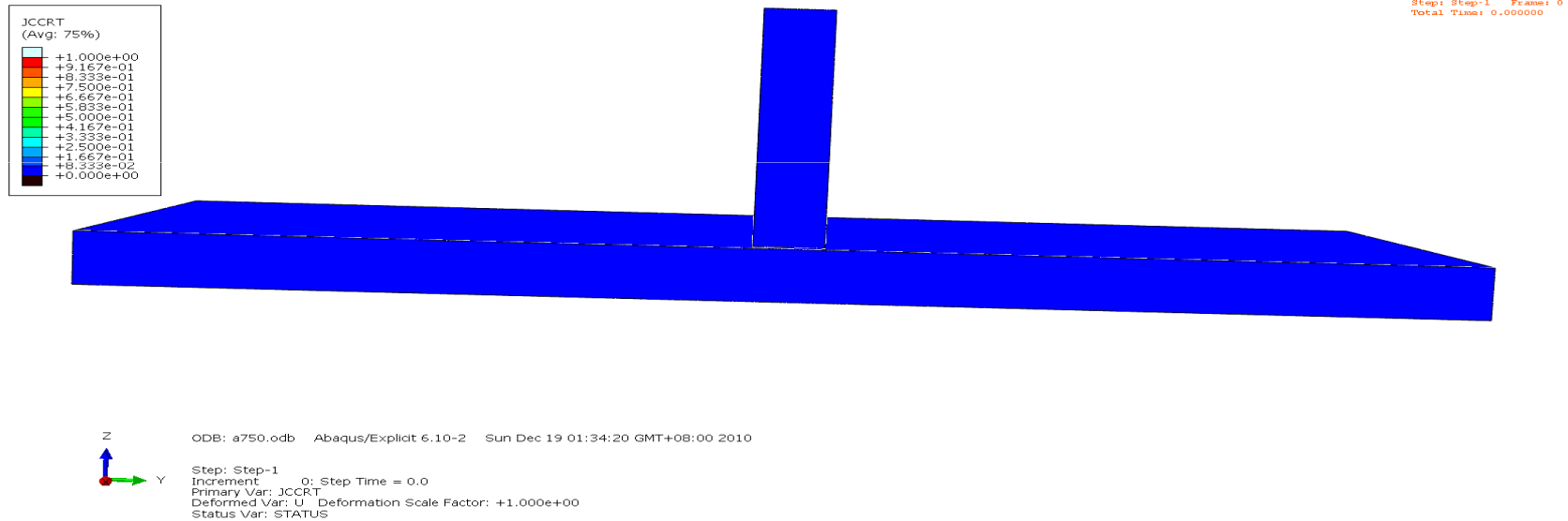
70μs时刻

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we can change the world.

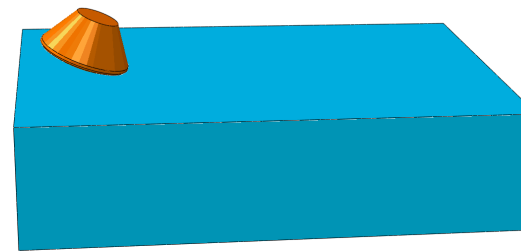
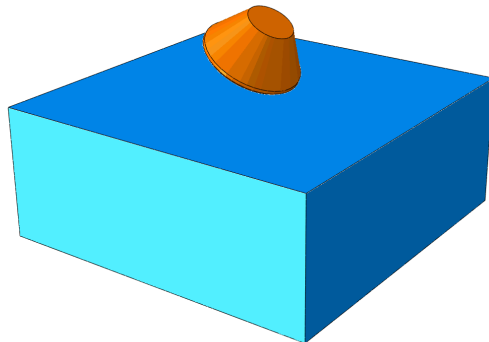
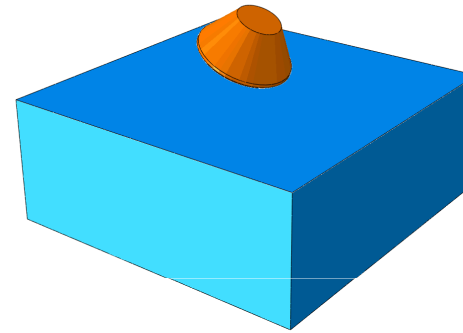
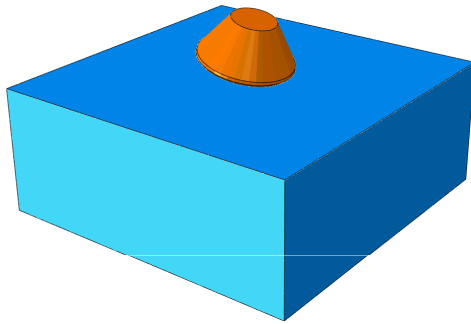
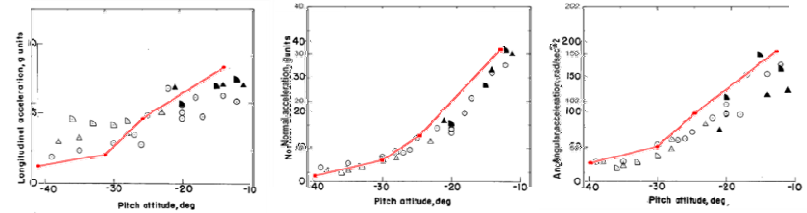
Abaqus模拟A3钢钝头弹撞击45钢

•以750m/s速度撞击



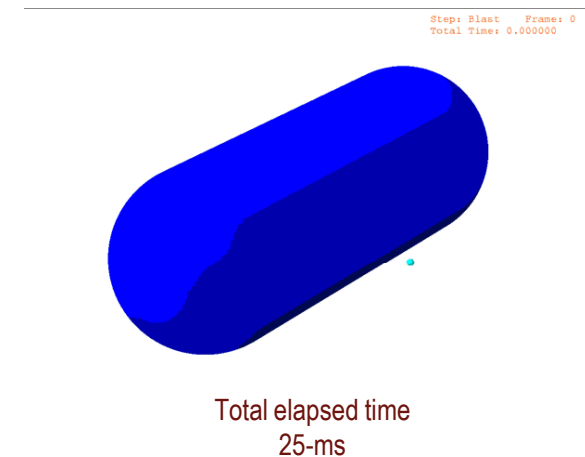
航天器返回舱水上着陆分析

CEL (航天器返回舱水上着陆分析)

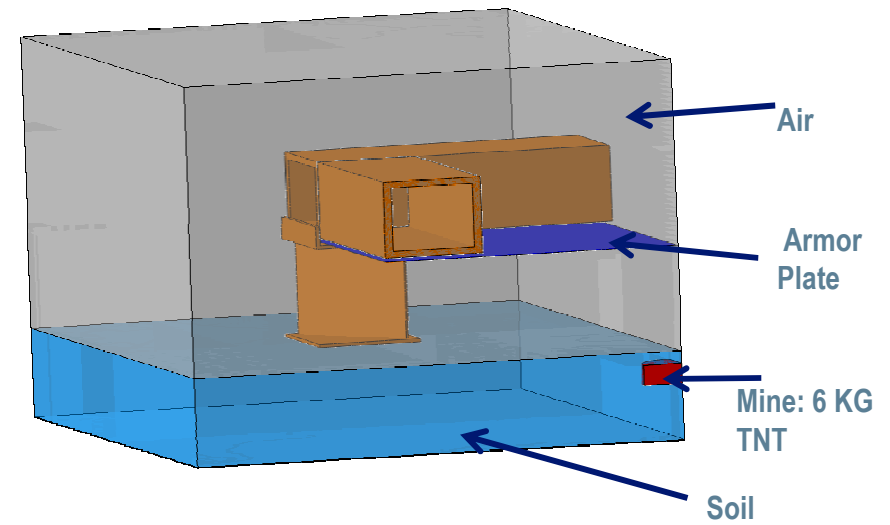
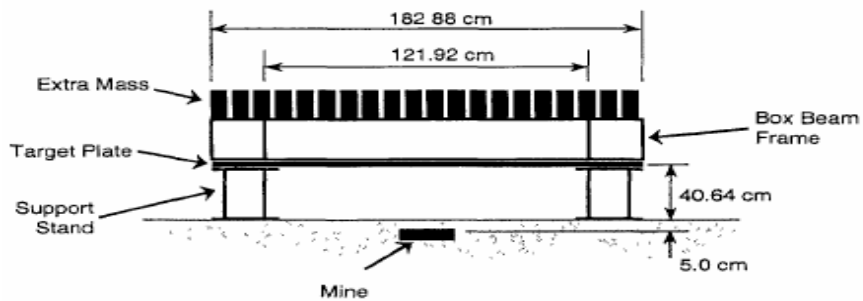


CEL模拟水下爆炸

- ▶ CEL approach effectively captures key results
 - ▶ Load from the blast on the shell not sufficient to breach
 - ▶ Blast is however sufficient to cause large plastic deformation
 - ▶ Bulk translation of the cylinder through the water is evident
 - ▶ Good symmetry above/below CL evident – even with coarse fluid model



地雷爆炸冲击



Quarter-symmetric CEL model

Experimental setup

地雷爆炸冲击

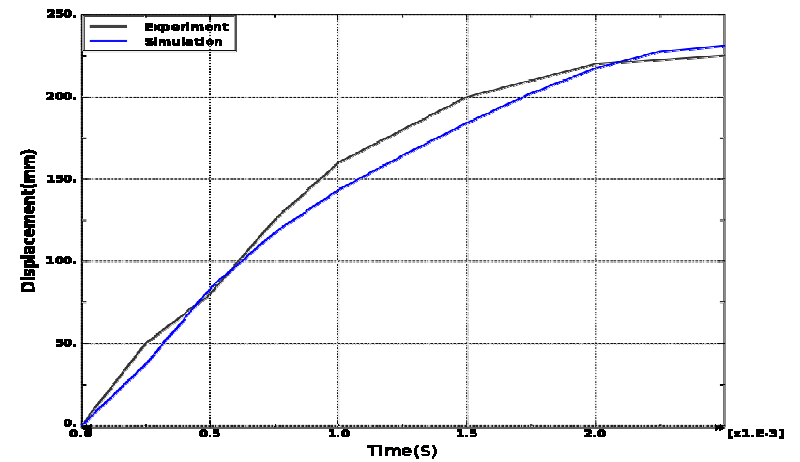
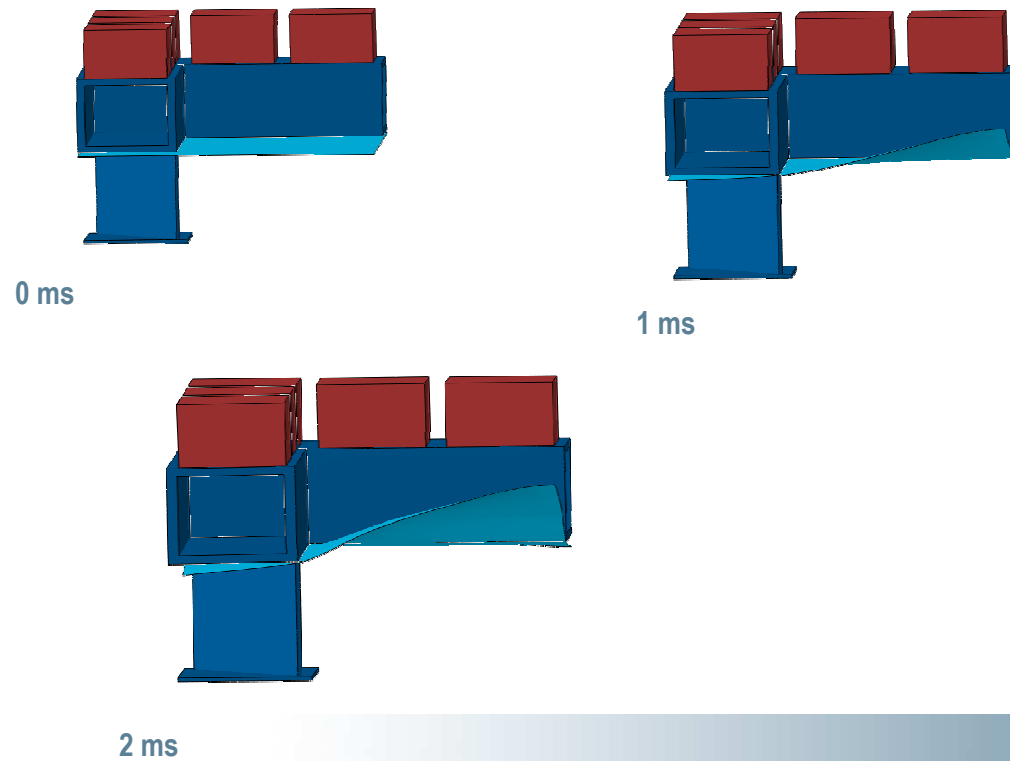
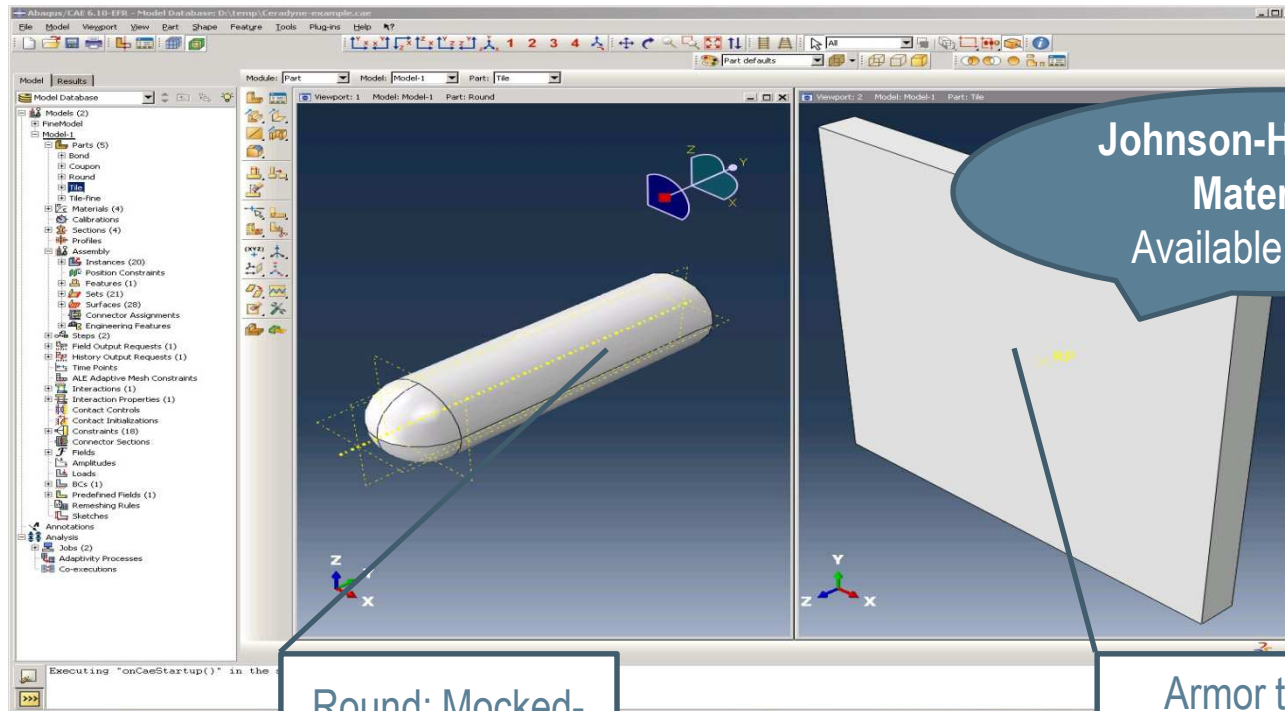
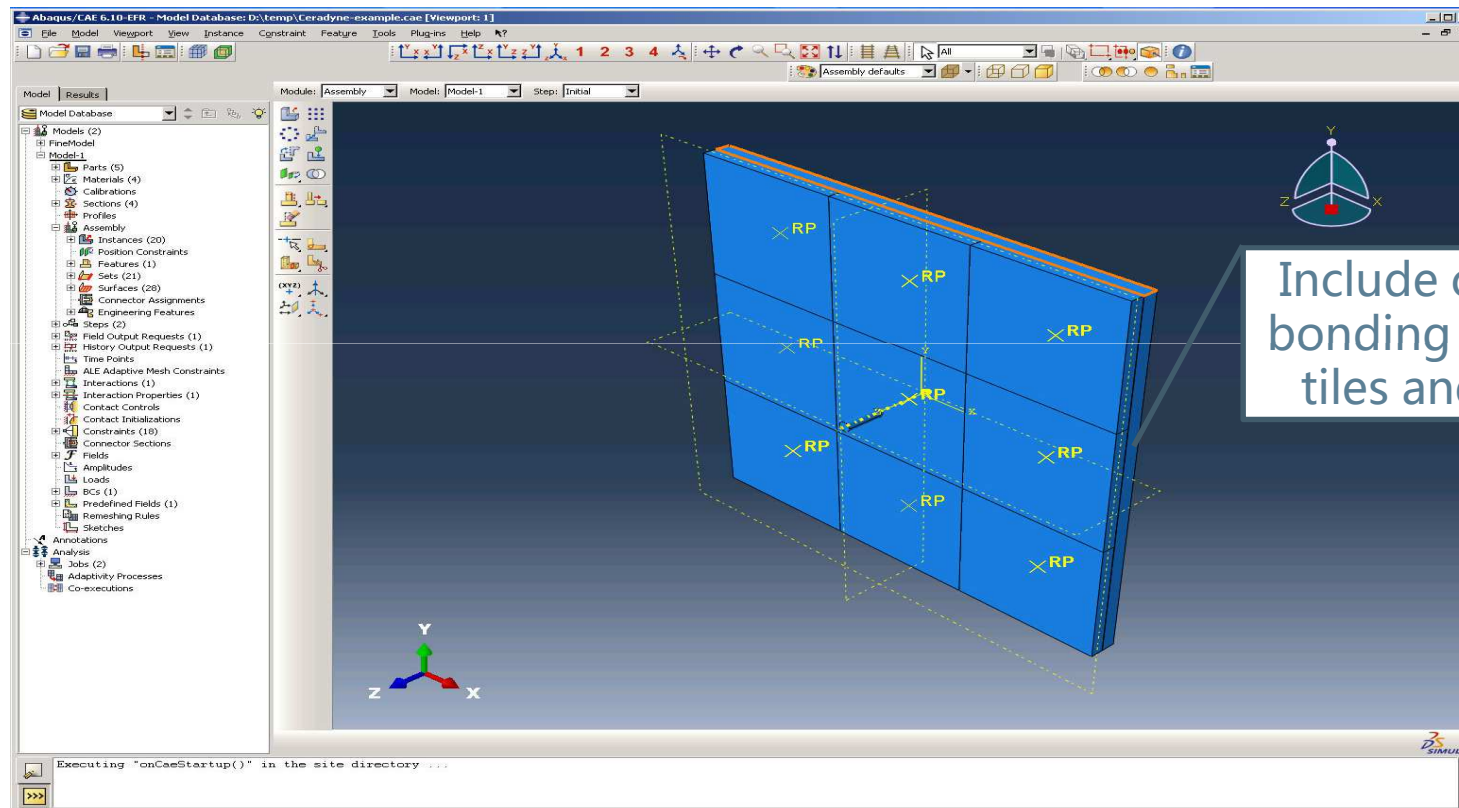


Plate Deformation: Test vs Simulation

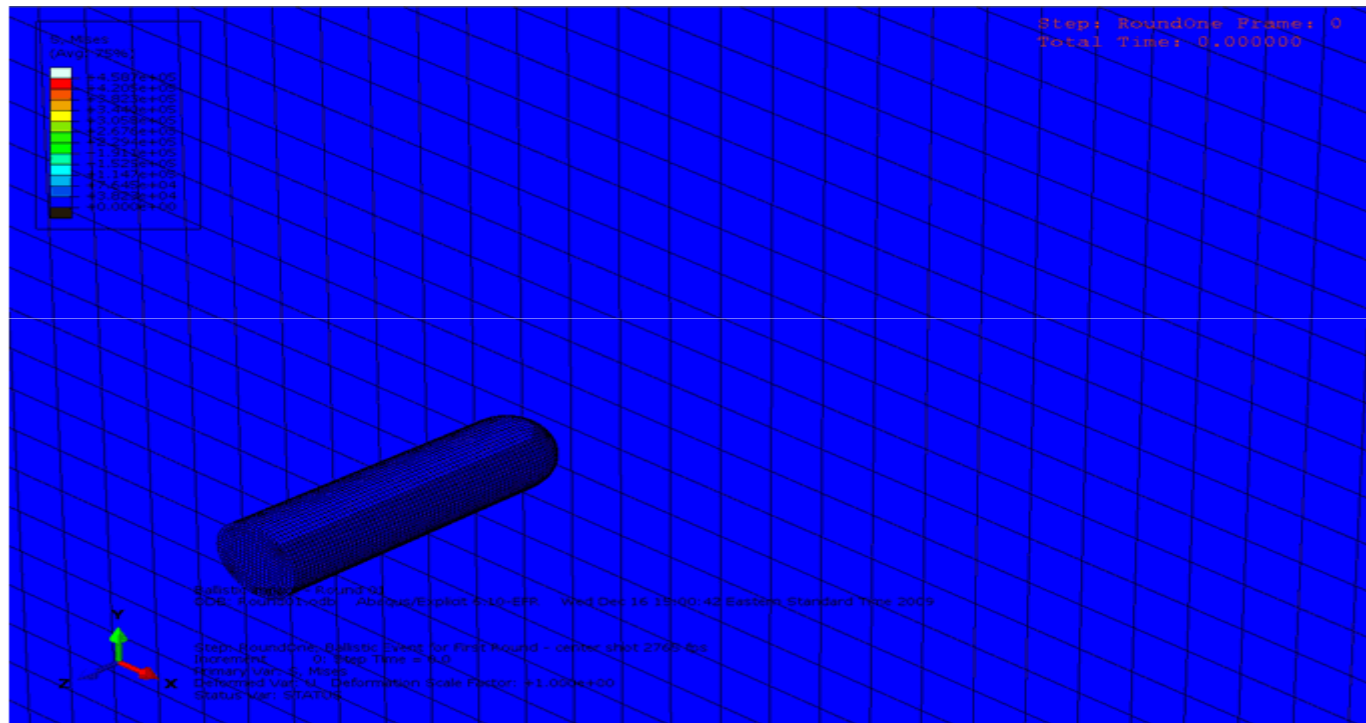
Case Study: Ballistic Test Panel



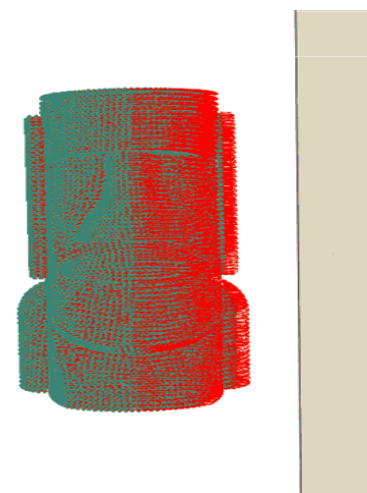
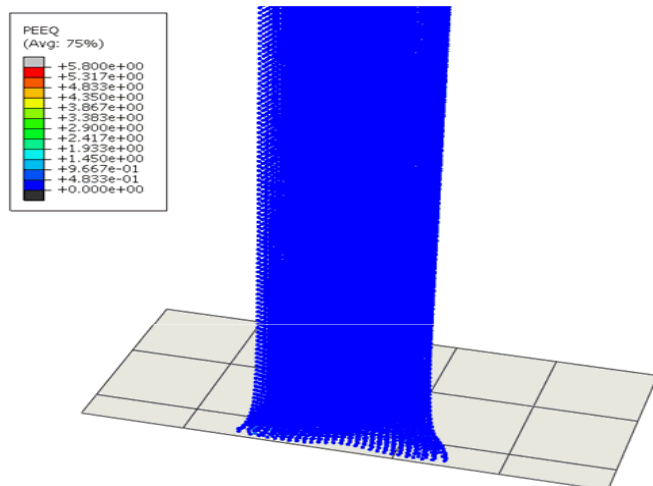
Case Study: Ballistic Test Panel



Case Study: Ballistic Test Panel



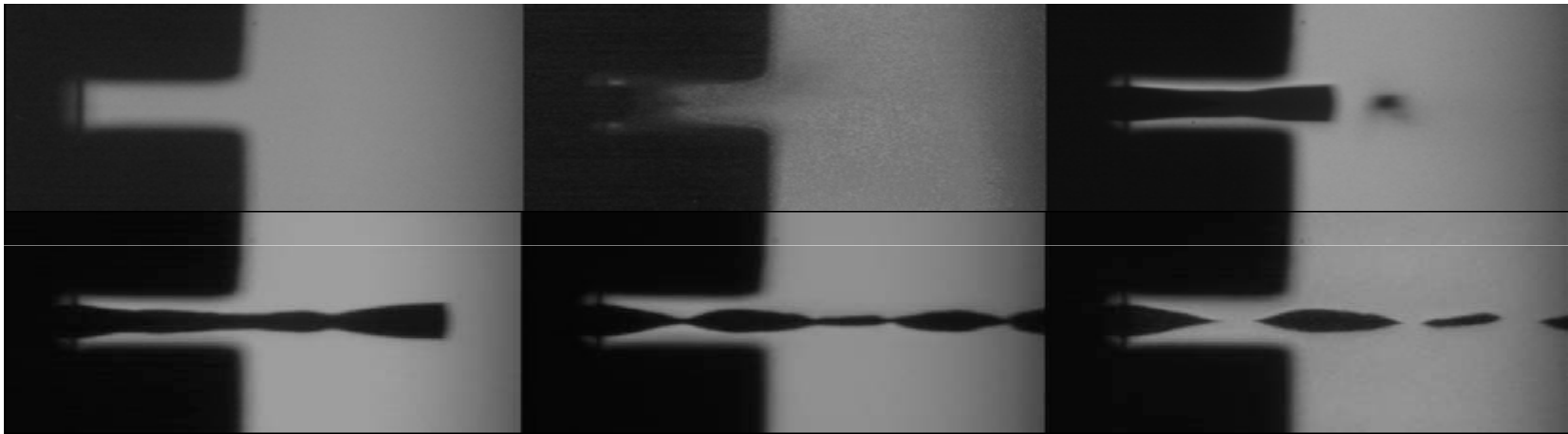
SPH



Dynamic Tensile Extrusion of Copper

"Validation of Abaqus Explicit – CEL for classes of problems of interest to the U.S. Army

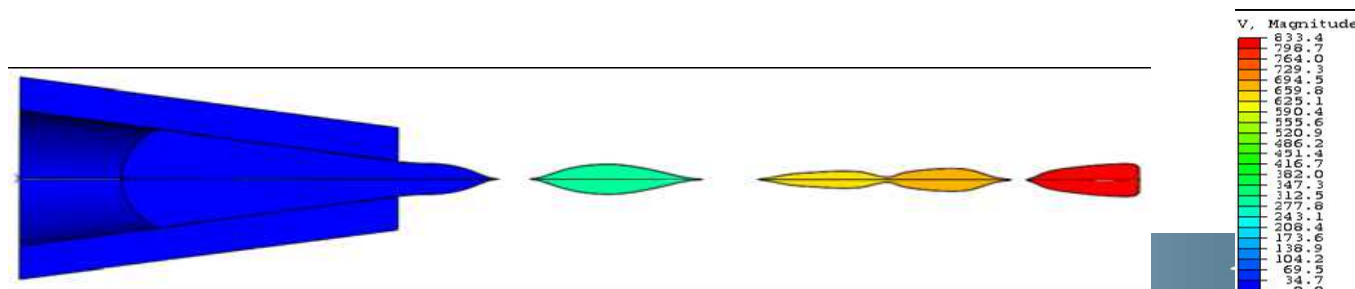
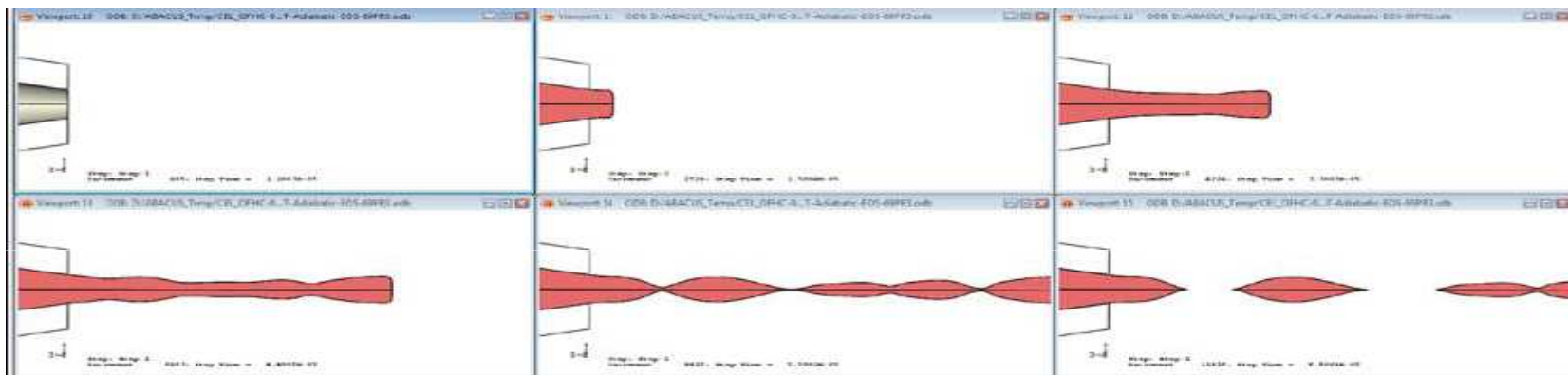
"Picatinny Arsenal – 2010 SCC



Dynamic Tensile Extrusion of Copper

"Validation of Abaqus Explicit – CEL for classes of problems of interest to the U.S. Army

" Picatinny Arsenal – 2010 SCC



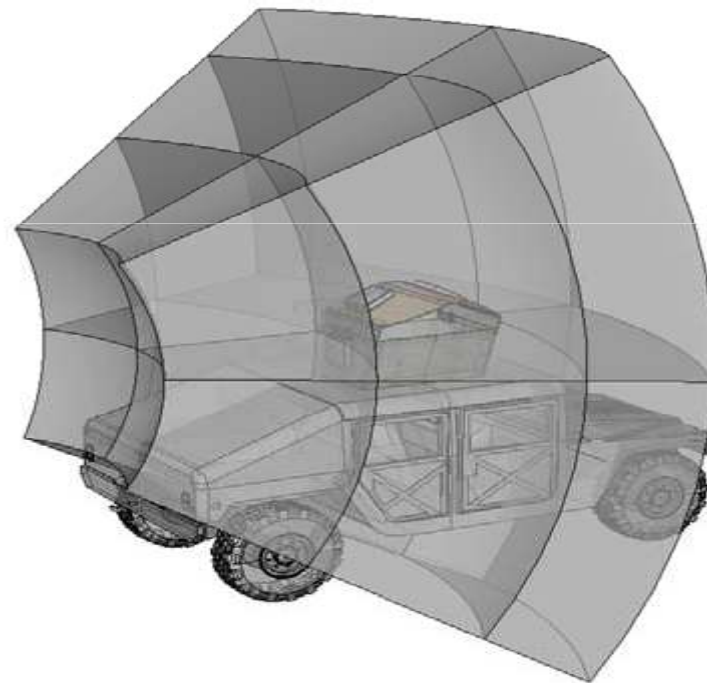
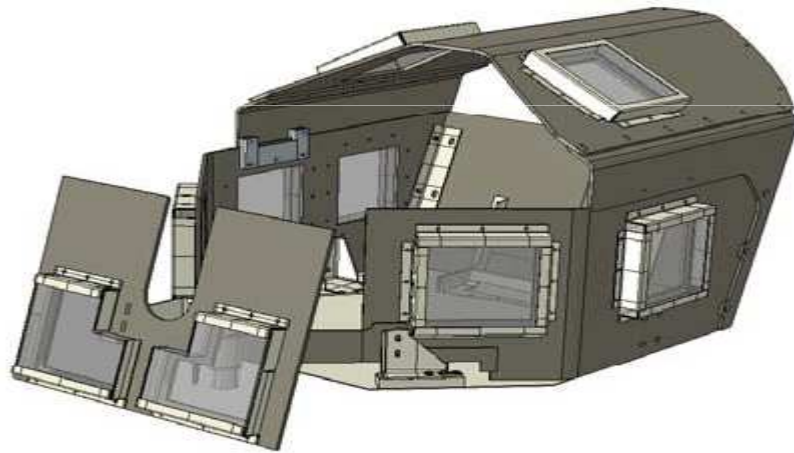
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US Army

“Novel Approach to Conducting Blast Load Analyses Using Abaqus/Explicit-CEL” Picatinny Arsenal – 2010 SCC

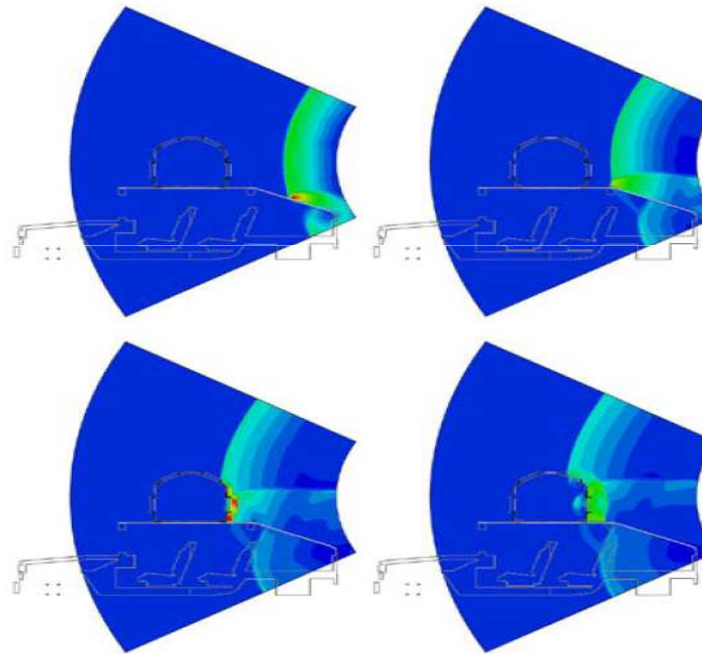
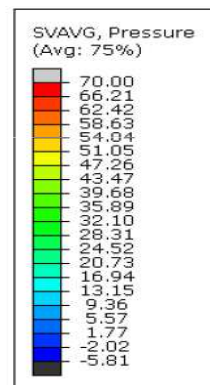
► 模拟悍马车的炮手保护装置在爆炸波的响应



US Army

“Novel Approach to Conducting Blast Load Analyses Using Abaqus/Explicit-CEL” Picatinny Arsenal – 2010 SCC

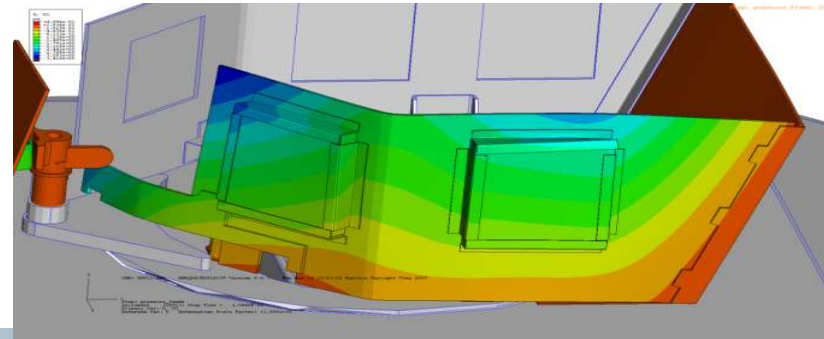
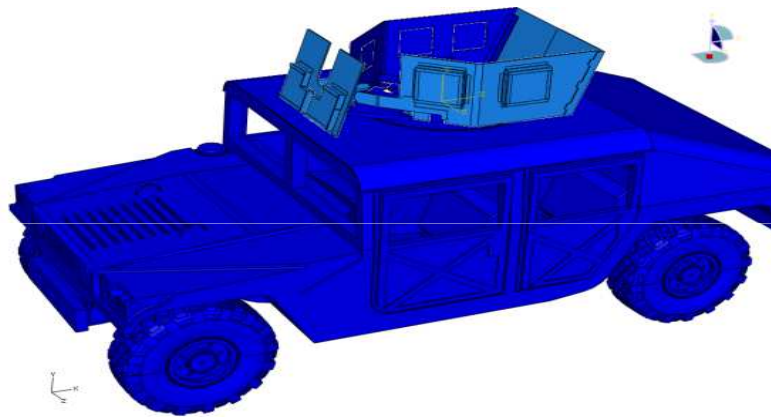
► 爆炸波的压力分布



US Army

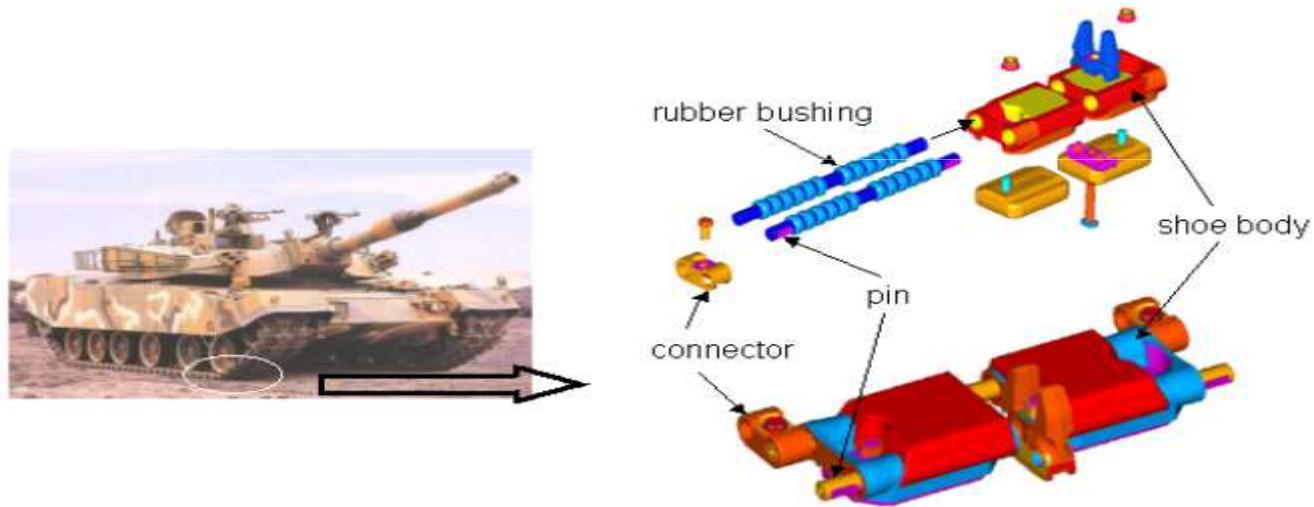
“Novel Approach to Conducting Blast Load Analyses Using Abaqus/Explicit-CEL” Picatinny Arsenal – 2010 SCC

► 结构位移变形



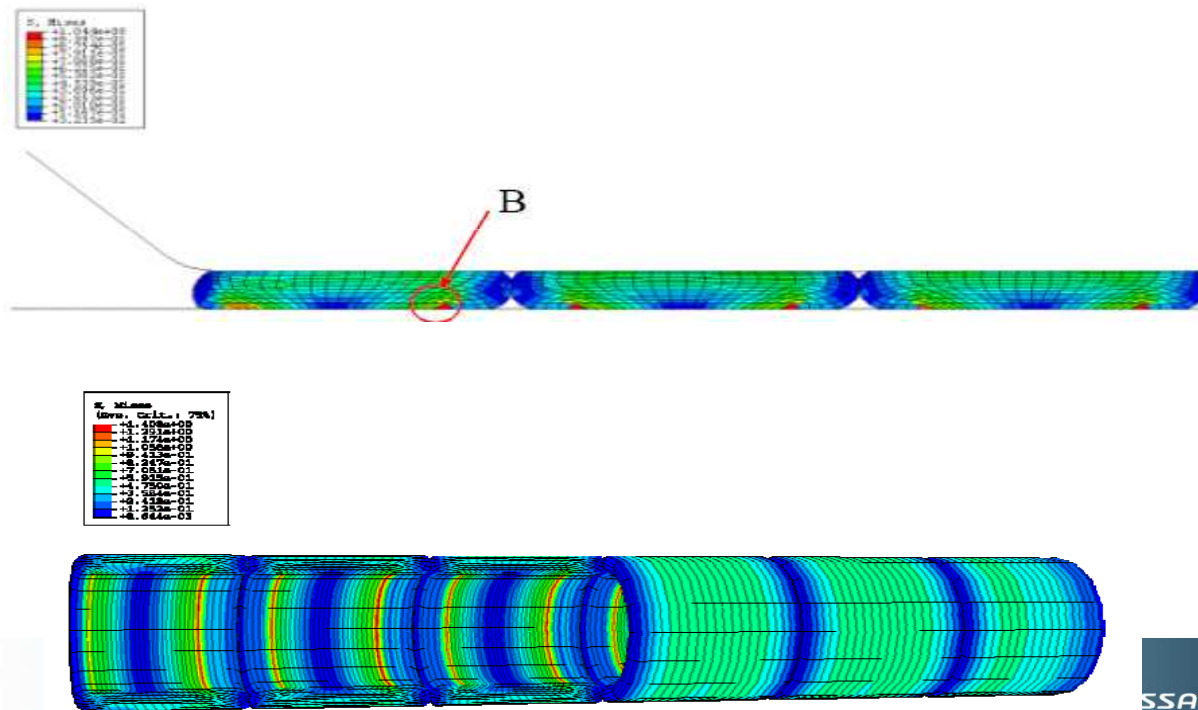
坦克履带胶环分析

- ▶ 韩国LG Cable公司采用Abaqus进行坦克履带胶环的装配应力和工作应力的分析



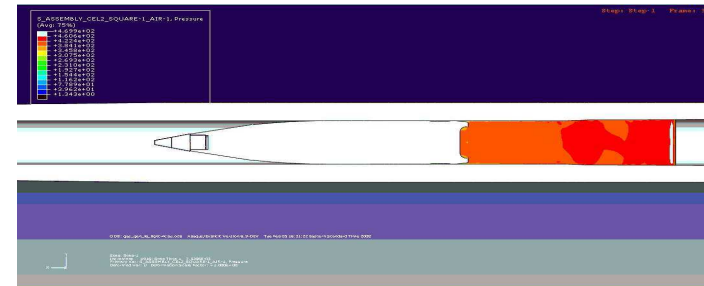
坦克履带胶环分析

- ▶ 模拟装配压缩载荷（装配中插入track shoe）；各种工况下载荷如联合的压缩和扭转载荷（履带拉伸和相对track shoe旋转）

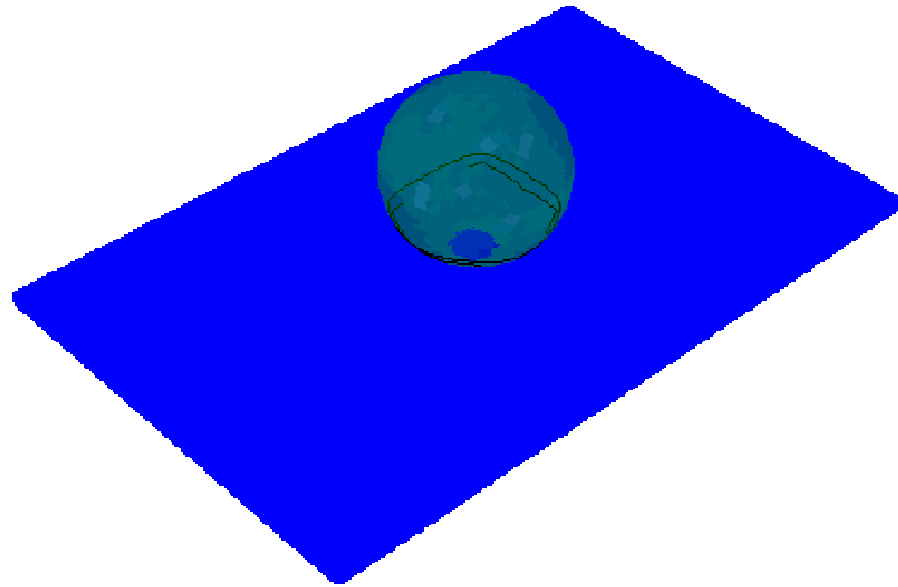


内弹道分析

- ▶ 坦克大炮等在炮弹发射时，内弹道的受力情况比较复杂，需要对结构各个部分在遭受高压气体冲击时的强度进行分析；炮弹和炮膛之间的摩擦分析，考察磨擦产生的热量以及对弹道的磨损；炮弹出膛时的速度和姿态；炮弹出膛瞬间对炮筒及支座的冲击。

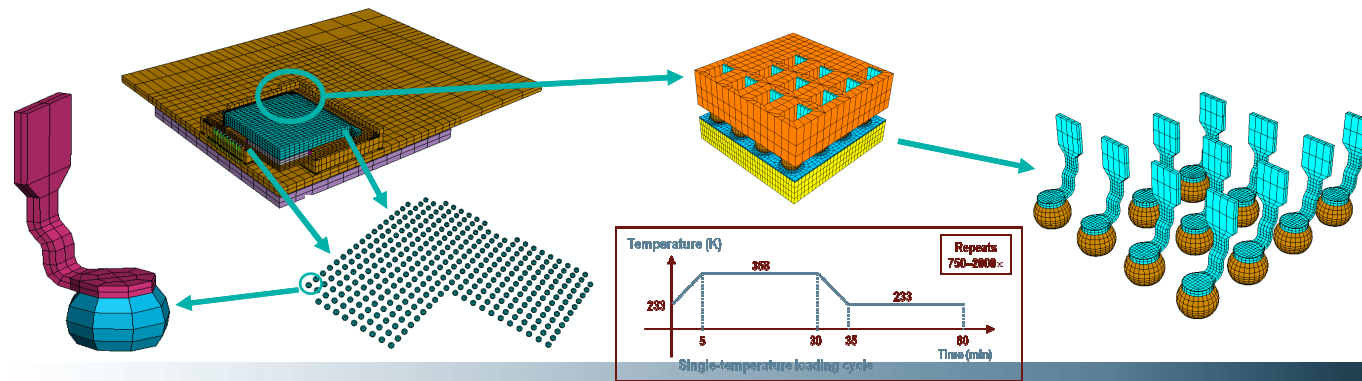


玻璃屏幕碰撞分析



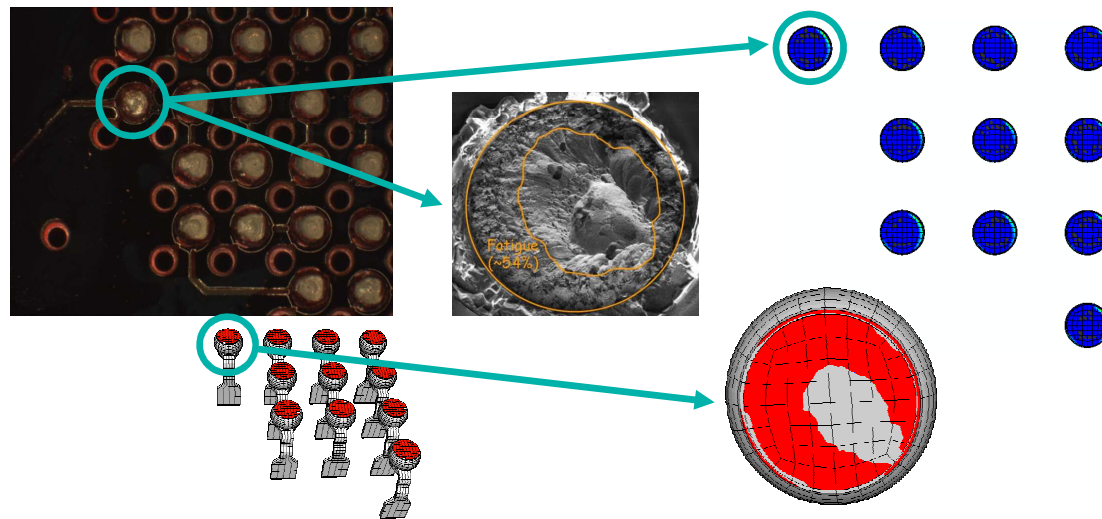
焊球疲劳开裂

- ▶ “Use Direct Cyclic and Sub-Modeling Approach to Analyze Solder Joint Fatigue Reliability,” Intel, SIMULIA – ASME-IMECE 2007
 - ▶ BGA solder joint fatigue reliability study for LGA sockets
 - ▶ Mechanical loading prior to cyclic thermal loading
 - ▶ Global-sub-modeling approach is used to predict the critical location



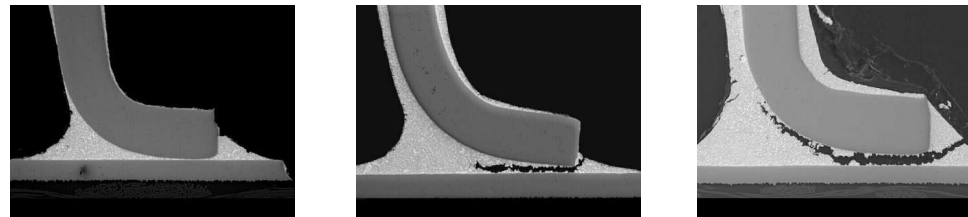
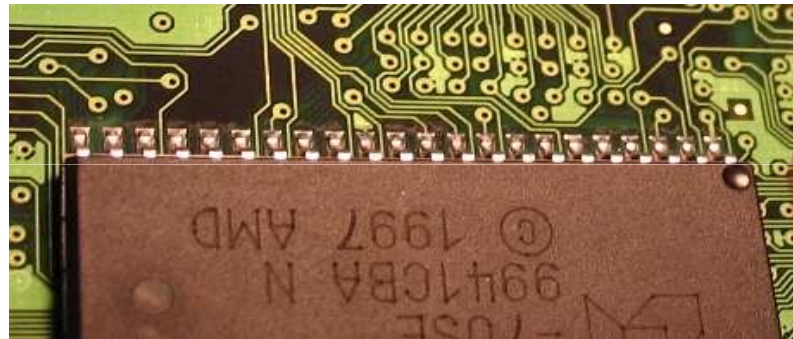
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- ▶ “Use Direct Cyclic and Sub-Modeling Approach to Analyze Solder Joint Fatigue Reliability,” Intel, SIMULIA – ASME-IMECE 2007
- ▶ BGA solder joint fatigue reliability study for LGA sockets



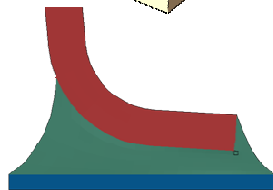
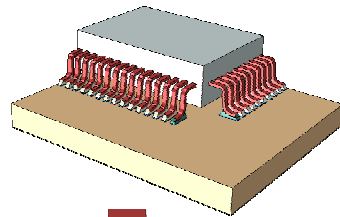
管脚疲劳开裂

- ▶ INSIGHTS Oct-2007, “It’s all in the Solder Joints”
 - ▶ Solder joint reliability analysis of automotive electronics under cyclic thermal loading

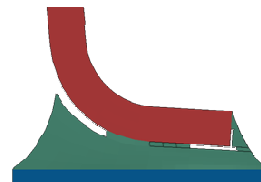
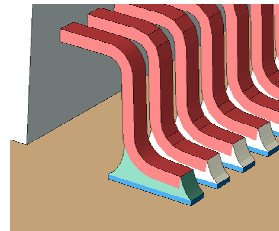


管脚疲劳开裂

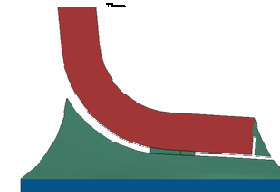
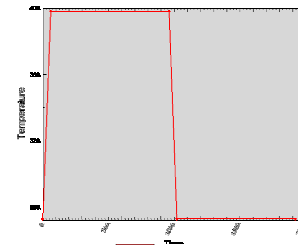
- Tech Brief: Low-cycle Thermal Fatigue of a Surface-Mounted Electronics Assembly
 - Solder material (63Sn/37Pb) modeled using temperature-dependent elasticity, power-law creep



Damage initiation at joint toe
Cycle number 199



Damage evolution
Cycle number 749



Damage evolution
Cycle number 801

管脚疲劳开裂

- ▶ Tech Brief: Low-cycle Thermal Fatigue of a Surface-Mounted Electronics Assembly
 - ▶ Solder material (63Sn/37Pb) modeled using temperature-dependent elasticity, power-law creep

